

Course Code	21CSC205P	Course Name	DATABASE MANAGEMENT SYSTEMS		Course Category	C	PROFESSIONAL CORE				L	T	P	C
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Pre-requisite Courses	Nil	Co- requisite Courses	Nil	Progressive Courses	Nil									
Course Offering Department	School of Computing		Data Book / Codes / Standards		Nil									

Course Learning Rationale (CLR):		The purpose of learning this course is to:
CLR-1:	understand the fundamentals and need of Database systems, Architecture, Languages	
CLR-2:	conceive database design through Relational model, Relational Algebra	
CLR-3:	design Logical schema with constraints, Familiarize SQL Queries	
CLR-4:	standardization of Database through Normalization	
CLR-5:	understand Storage Management, the practical problems of Concurrency control, Failures and recovery, NoSQL database	

Course Outcomes (CO):		At the end of this course, learners will be able to:
CO-1:	acquire knowledge on DBMS architecture and languages	
CO-2:	acquire knowledge on Relational languages and design a database	
CO-3:	implement the Database structure with SQL	
CO-4:	removal of anomalies using Normalization concepts	
CO-5:	visualizing storage structure, handling concurrency, Failure and recovery principles, NoSQL concept	

Unit-1 - Introduction	Issues in File Processing System, Need for DBMS, Basic terminologies of Database, Database system Architecture, Various Data models, ER diagram basics and extensions, Case study: Construction of Database design using Entity Relationship diagram for an application such as University Database, Banking System, Information System										12 Hour			
Unit-2 - Relational DBMS	Conversion of ER model to Relational Table, Case study: Apply conversion concept. Discussion of various design issues. Pitfalls in Relational Database systems, Understanding various Relational languages such as Tuple Relational calculus, Domain relational calculus, Calculus Vs Algebra, Computational capabilities. Case Study: Applying Relational Algebra for all the queries of application Designed.										12 Hour			
Unit-3 – SQL	SQL commands, Constraints, Joins, set operations, Sub queries, Views, PL – SQL, Triggers, and Cursors. Case Study: Implement all the queries using SQL, PL-SQL, Cursor and Triggers										12 Hour			
Unit-4 - Normalization	Normalization, Need for Normalization, NF1, NF2, NF3, NF4, NF5. Case study: Apply Conversion rules and normalize the Database										12 Hour			
Unit-5 – Concurrency Control	Storage Structure, Transaction control, Concurrency control algorithms, Issues in Concurrent execution, Failures and Recovery algorithms Case study: Demonstration of Entire project by applying all the concepts learnt with minimum Front end requirements, NoSQL Databases-Documents Oriented, Key value pairs, Column Oriented and Graph										12 Hour			

Learning Resources	1. Abraham Silberschatz, Henry F. Korth, S. Sudharshan, Database System Conceptsll, Seventh Edition, Tata McGraw Hill, 2019. 2. Ramez Elmasri, Shamkant B. Navathe, Fundamentals of Database Systemsll, Sixth Edition, Pearson Education, 2011. 3. C.J Date, A Kannan, S Swamynathan, An Introduction to Database Systems, Eight Edition, Pearson Education, 2006.	4. RaghuramaKrishnan, Johannes Gehrke, Database Management Systems, 3rdEdition, McGrawHill Education, 2003. 5. Principles of Database Systems, J.D. Ullman, Galgoli, 1982 6. NoSQL Dislilled, A brief guide to the emerging world of Polygot persistence, First Edition, Promod J,Sadalage Martin Fowler, 2012
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Learning Assessment		Continuous Learning Assessment (CLA)					
	Bloom's Level of Thinking	Formative		Project Based Learning		Report and Viva Voce	
		CLA-1 Average of unit test (20%)		CLA-2 (60%)		Final Examination (0% weightage)	
		Theory	Practice	Theory	Practice	Theory	Practice
Level 1	Remember	20%	-	-	-	-	-
Level 2	Understand	40%	-	-	-	-	-
Level 3	Apply	40%	-	-	-	-	-
Level 4	Analyze	-	-	-	30%	-	-
Level 5	Evaluate	-	-	-	30%	-	-
Level 6	Create	-	-	-	-	50%	-
	Total	100 %		100 %		100%	

Course Designers	
Experts from Industry	Internal Experts
1. Ms.Sangeetha Jayaprakash, Database Architect, BOSCH India	1. Dr.M.Thennmozhi, SRMIST
2. Dr.Manipoonchelvi, Senior Technical Manager, HCL Technologies	2. Ms.K.Srividya, SRMIST